A SHEAF MAPPING REPRESENTABLY WITH EFFECTIVE DESCENT TO AN ALGEBRAIC SPACE IS AN ALGEBRAIC SPACE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof contains a multi-line align and a bundle-like display; both are part of the counted proof span and are preserved verbatim.

Editorial note. The five references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-spaces/retained-preamble.tex, sliced mechanically from the chapter's own preamble.tex.

Allowed standard prerequisites (editorial).

- Effective descent for morphisms of a type of property P. Descent, Definition descent-definition-descending-types-morphisms, used at proof lines 2010-2013 and 2051-2055. Is the hypothesis that makes the descent datum representable.
- Descent data for sheaves are effective. Descent, Lemma descent-lemma-descent-data-sheaves, used at proof lines 2043-2050. Identifies the descent datum with the one attached to the sheaf G x_F T.
- A representable transformation over a fixed algebraic space has algebraic space source. Lemma lemma-representable-over-space (1926-1936), used at proof lines 2057-2059. Concludes that G is an algebraic space.
- Properties of representable morphisms of spaces. Lemma lemma-representable-morphisms-spaces-property, used at proof lines 2059-2061. Transfers property P to G -> F.

Source reference. Original definition 02W2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3789–3789. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original definition 023V: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2756–2756. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02W5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3792–3792. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02WZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3822–3822. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02WY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3821–3821. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme contained in Sch_{fppf}. Let $G \to F$ be a transformation of presheaves on $(Sch/S)_{fppf}$. Let $\mathcal{P}$ be a property of morphisms of schemes. Assume*

1. *$\mathcal{P}$ is preserved under any base change, fppf local on the base, and morphisms of type $\mathcal{P}$ satisfy descent for fppf coverings, see Descent, Definition 02W2,*
2. *G is a sheaf,*
3. *F is an algebraic space,*
4. *there exists a $U \in \mathrm{Ob}((Sch/S)_{fppf})$ and a surjective étale morphism $U \to F$ such that $V = G \times_F U$ is representable, and*
5. *$V \to U$ has $\mathcal{P}$.*

Then G is an algebraic space, $G \to F$ is representable and has property $\mathcal{P}$.

Proof. Let T be a scheme and let $T \to F$ be a morphism. Then $U \times_F T \to T$ is surjective étale, hence $\{U \times_F T \to T\}$ is a covering for the étale topology. Consider

$$W = G \times_F (U \times_F T) = V \times_F T = V \times_U (U \times_F T).$$

It is a scheme since F is an algebraic space. The morphism $W \to U \times_F T$ has property $\mathcal{P}$ since it is a base change of $V \to U$. There is an isomorphism

$$\begin{aligned} W \times_T (U \times_F T) &= (G \times_F (U \times_F T)) \times_T (U \times_F T) \\ &= (U \times_F T) \times_T (G \times_F (U \times_F T)) \\ &= (U \times_F T) \times_T W \end{aligned}$$

over $(U \times_F T) \times_T (U \times_F T)$. The middle equality maps $((g, (u_1, t)), (u_2, t))$ to $((u_1, t), (g, (u_2, t)))$. This defines a descent datum for $W/U \times_F T/T$, see Descent, Definition 023V. This follows from Descent, Lemma 02W5. Namely we have a sheaf $G \times_F T$, whose base change to $U \times_F T$ is represented by W and the isomorphism above is the one from the proof of Descent, Lemma 02W5. By assumption on $\mathcal{P}$ the descent datum above is representable. Hence by the last statement of Descent, Lemma 02W5 we see that $G \times_F T$ is representable. This proves that $G \to F$ is a representable transformation of functors.

As $G \to F$ is representable, we see that G is an algebraic space by Lemma 02WY. The fact that $G \to F$ has property $\mathcal{P}$ now follows from Lemma 02WZ. □